Hardware User Manual

AnyMotion Series

PCI-N404

AJINEXTEK AXT

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Home Page : www.ajinextek.com
E-mail : support@ajinextek.com

Useful Contact Information

Customer Support Seoul
Tel : 82-31-436-2180~2 Fax: 82-31-436-2183

Customer Support Taegu
Tel : 82-53-593-3700~2 Fax: 82-53-593-3703

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Revision History

Manual	PCB	Library	Comments
Rev. 1.0 issue 1.0	Rev. B2.1	AXL1.0.0.0 and above	DEC., 2006.
Rev. 1.0 issue 1.0	Rev. B2.2	AXL3.1.7.0 and above	JUL., 2008.

	This is a warning sign that requires your attention. Ignoring the instructions associated with this sign may result the damages and the malfunctions of board

	This sign is for the reference articles and information for the usage of the board.

1. Overview

1.1. Introduction

This product is a PCI Half size board for four axes pulse output and the board is designed and manufactured for the easy and simple realization of the high speed and precision motion control. The motion control algorithm processing is done by CAMC-QI (motion control chip) installed on each product. This on-board custom LSI chip dedicated for the motion control processes complex motion control algorithms like position control and velocity control including acceleration and deceleration in connection with the Ajinextek's complete software package. The PCI-N404 is a single PCI bus type board that can be applied to the various industry standard buses and stand alone control system for the precision control of step motors, DC servo motors, AC servo motor that are used as the main actuators of the industrial equipments including semiconductor manufacturing equipments, textile equipments, packing equipments, PCB cutting machines, and so on.

1.2. Application

This board can be applied to building the precision motion control system including but not limited to the following applications that use step motors or AC/DC servo motors.

▶ Semiconductor manufacturing equipment
▶ Motion Vision
▶ Robot engineering
▶ X-Y table
▶ Winding machine
▶ Embroidery machine
▶ X-Y position control
▶ Loading/Unloading
▶ Step motor control
▶ Equipment control
▶ Roll Feeding
▶ Pick & Placing
▶ Research & development

1.3. Functions and Features

▶ Maximum number of control axis

Four axes motion control with the on board dedicated motion control chip (CAMC_QI)

▶ Interpolation drive

User programmable three axes linear and two axes circular interpolation processed per selected motion control chip

▶ Drive methods

Drive by specified number of pulses, continuous drive, signal search drive, signal detection-2, interpolation drive, MPG drive, etc

▶ Bus interface

PCI, Memory access, 16bit access, PCI 2.1

▶ Encoder input

Single-phase pulse/up-down pulse input available

Two-phase pulse: 1, 2, or 4 multiplication settings available

▶ Pulse output method specification

Total ten pulse types including four of one pulse types, four of two pulse types, and two of two-phase pulse types

▶ Override function

Velocity and position override functions while performing the programmed motion

▶ Acceleration/deceleration mode

Supported velocity profiles are symmetric trapezoid, symmetric S-curve, asymmetric trapezoid, and asymmetric S-curve

▶ Pulse output accuracy

Pulse output error less than 0.1% from the specified values

▶ Pulse output rate

~ 10.000Mpps

▶ Pulse output interface

Interface provided by the linear driver IC (74ALS174)

▶ Encoder input interface

Line receiver (26LS32) input

▶ Encoder input requirements

~ 4㎒(pulse count rate : 16Mcount/Sec)

▶ Read/writable 28bit counter for encoder and internal counter

▶ Functions to stop the motion in response to the external soft stop and emergency stop signal

▶ Input/output signal for servo motor

ALM(alarm signal), INPO(positioning decision signal), SVON(servo-on signal), CLR(remaining pulse deletion signal), ALMC(servo alarm removal signal)

▶ 40 universal input/output signals

Five input and five output signals for each axis (multiplexed with the Z-phase of encoder, SVON, and ALMC)

▶ Each output driver contact supports up to 100mA.

Two high-output driver contacts for each axis (8 in total)

▶ Script functions and functions to capture the status of registers

Capability of continuous drive by executing the commands stored in the chip triggered by specific events.

Capability to capture the status of internal registers of the chip when triggered by specific events

▶ Motion drive by external two-phase signals

1:N, N:1 drive for MPG(Manual Pulse Generation) available

▶ Capability to check the status of motion

LED displays to check the status of the motion during pulse output (limited to Z axis only)

1.4. Requirements

Table 1. Electrical and Environmental Requirements of PCI-N404 module

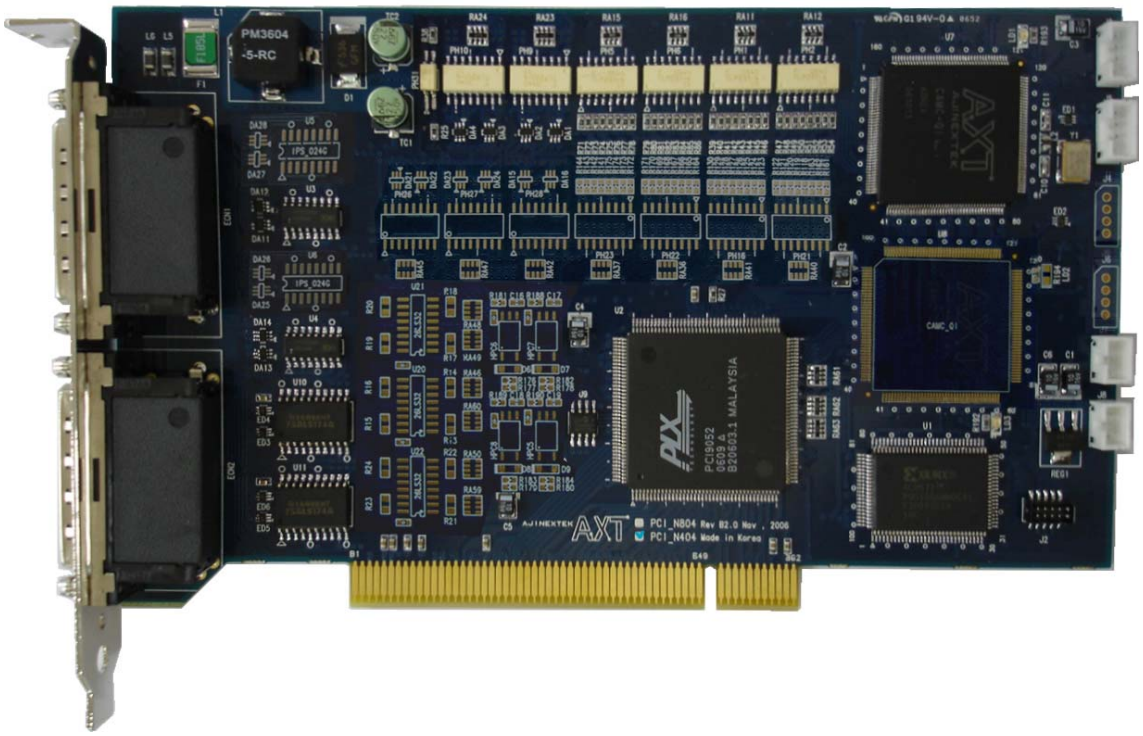
Items	Specification
Motion Control Chipset	CAMC-QI 1EA
Controls Axis	4 Axes
Pulse Output	~ 10.000Mpps, support two phase out method
BUS Interface	PCI 2.1, Memory access,
Encoder Input	~ 4㎒(16Mcount/Sec)
Encoder Counter	28-Bit
UIN/OUT	20EA/20EA (16EA Power FET output[100mA maximum])
Operating Temperature Rang	0 ~ +60℃
Operating Humidity Range	80% or below
Board Size	174.63 × 106 mm
Print Panel Connector	Two of 68Pin CHANP connectors in two rows
Other	100mA of maximum input/output current for each external connector

Operation Voltage:	
5V	Voltage Range : 4.7V to 5.3V

Maximum current consumption	
5V	1A

2. Module Structure

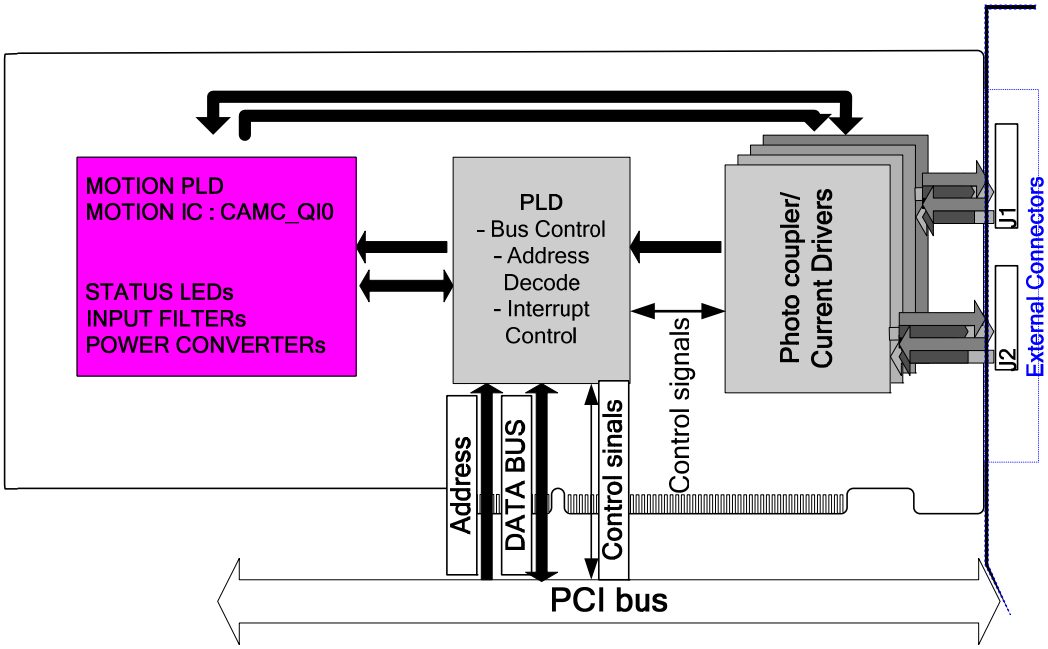
2.1. Board Layout



Picture 1. Actual Photo of PCI-N404

2.2. Hardware Block Diagram

Major blocks of the PCI-N404 are PLD block, CAMC-QA motion control chip, and input/output blocks for pulse output and contact signals. PLD block interfaces to the PCI bus and various control signals and provides the interfaces to the CAMC-QI motion control chip in addition to the various auxiliary functions and interrupt handling part. CAMC-QI motion chip processes various motion control signals including pulse output, encoder input, limit sensor input, and other signals related to the motion control. Motion control signals are electrically isolated from the external system by photo couplers. Line driver IC drives the output pulse signals for all output methods, and this interface makes the signals less susceptible to the external noise generated by the system operation.



Picture 2. Block Diagram of PCI-N404

3. Installation

3.1. Hardware Installation

After confirming the product components, the board must be installed in the following steps. The product package comprises PCI-N404 board and optional cables and terminal blocks.

① Turn off the IPC power and disconnect the power cord.

② Select the PCI slot to install the board.

③ Remove the panel on the empty slot to access the selected PCI slot.

④ Touch the metal parts of the case with bare hands to discharge any static electricity built up on clothes or body.

⑤ Insert the board into the selected PCI slot. Use screws to fix the brackets onto the IPC main body.

⑥ Confirm if it is properly installed.

⑦ Plug in the IPC power cord and turn on the power to start the system.

⑧ Check if the booting is completed successfully.

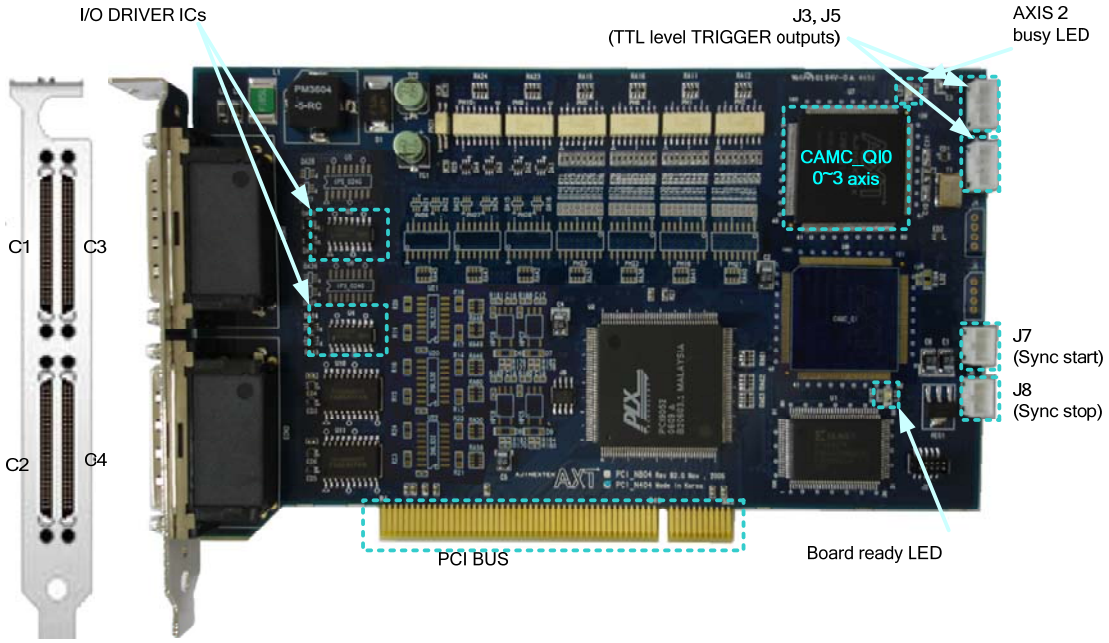
3.2. Software Installation

Refer to the EzConfig 2007 User's Manual for the installation of EzConfig software.

3.3. Hardware Details

3.3.1. Connectors and LEDs

(Connector 명칭이 이 그림에서만 C1~C4로 되어있음, 다른부분에서는 J1~J4)



Picture 3. PCI-N404 board and connector

Picture 3 shows the PCI-N404 board, external connectors and the indication LEDs.

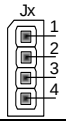
Each board includes LEDs (Board ready LED) indicating the operating status of the board, connectors to connect with external devices or terminal blocks with cables, and header jumpers to connect with internal TTL level trigger signals.

The C1 and C2 connectors for connecting with external devices are 68 pin CHAMP connectors that provide the connection for input/output sensors and driver pulse signals for each of axes. Refer to the board connection diagram for the detailed description of each pin.

C1 connector accommodates all signals related to the axis 0 and 1, and C2 connector accommodates all signals related to the axis 2 and 3. C3 and C4 are not connected

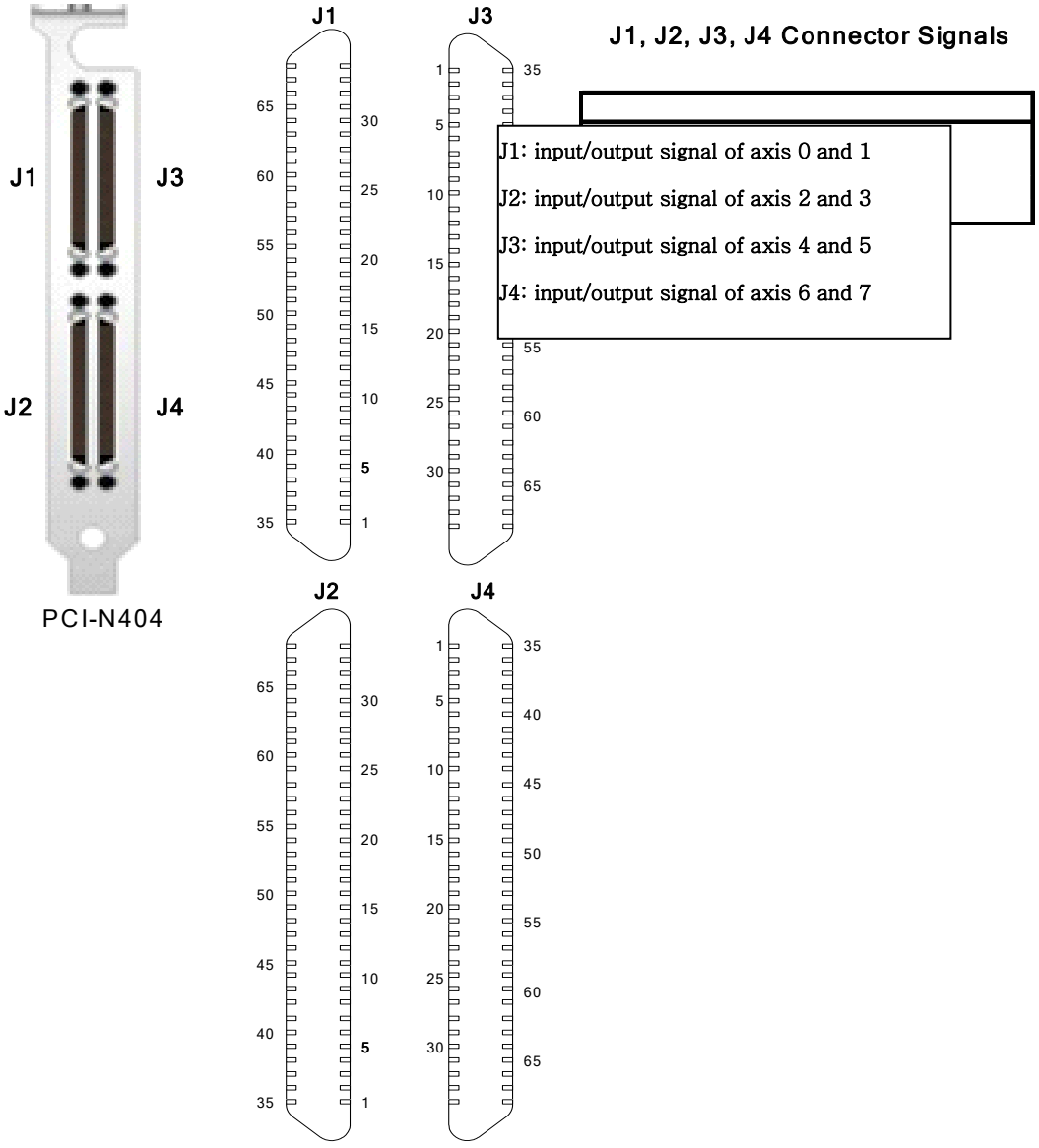
The board-ready LEDs are for checking the power supply status of the board and the status whether the board is operating in normal condition. Axis busy LED displays the operating status of CAMC-QI. The following table describes the functions of LEDs and jumpers.

PCB symbols	Function	Description
LD3	Status of board operation	Status of 5V supply, Status of board operation
LD1	CAMC_QI status	U7(CAMC-QI0) Z axis operating status Status of 1.8V power supply

J3(2mm pitch)  	Trigger signal (TTL) output connector	1 : TTL level (5V) trigger signal of axis 0 2 : GND 3 : TTL level (5V) trigger signal of axis 1 4 : GND
J5(2mm pitch)	Trigger signal (TTL) output connector	1 : TTL level(5V) trigger signal of axis 2 2 : GND 3 : TTL level(5V) trigger signal of axis 3 4 : GND
J7(2mm pitch)	Input/output connector board synchronization	1 : Sync Start group 1 input/output 2 : Sync Start group 2 input/output 3 : GND
J8(2mm pitch)	Input/output connector for board synchronization	1 : Sync Stop group 1 input/output 2 : Sync Stop group 2 input/output 3 : GND

3.3.2. Front Panel I/O Pin Array

The J1 and J2 CHAMP connectors shown in the following picture are for the input and output signals of the axis 0~1 and axis 2~3, respectively. J3 and J4 are not connected to any signals. Use the cables to connect with terminal blocks or external devices directly without terminal blocks.



Picture 4. CHAMP Connectors of PCI-N404 board (J1,J2)

3.4. Register

3.4.1. Register Map

The following table shows the local memory allocation details of the PCI-N404 board based on the base address assigned by the system.

Table 2. Full Address map of the PCI-N404 board

A12(MSB) ~ A0(LSB)

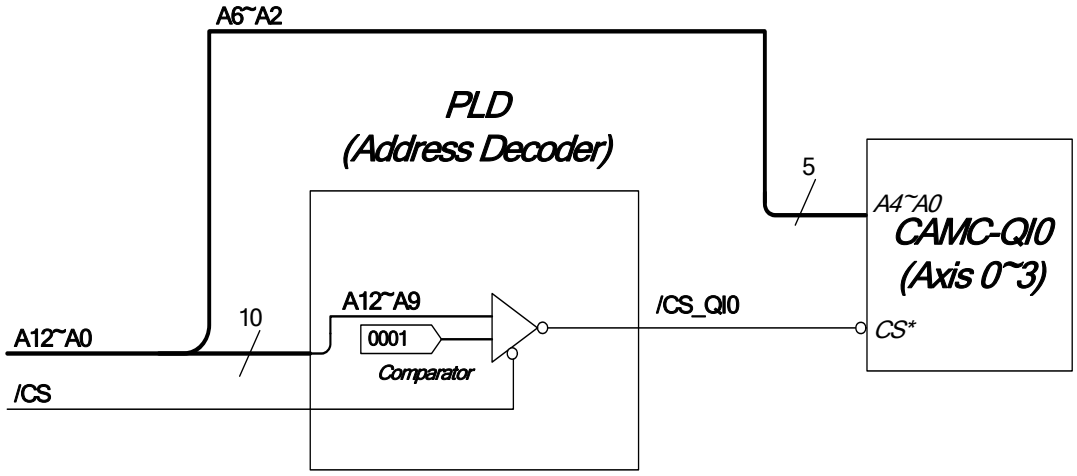
Region	Offset	PCI-N404 0000h – 1FFFh	Remarks
0000h – 03FFh	-	Address space for CAMC-QI0(U7) (AXIS0~3)	1Kbyte
0400h – 07FFh	-	Reserved	
0800h – 0BFFh	-	Reserved	
0C00h – 0FFFh	-	Reserved	
1800h – 1FFFh	-	Board Configuration Area	2Kbyte
	+000h	SYNC and Board Set	(1800h)
	+004h	Interrupt enable	(1804h)
	+006h	Interrupt flag	(1806h)
	+008h	Board LED	(1808h)
	+3F0h	Preamble : 0xB6	(1FF0h)
	+3F2h	Board ID(0x84)	(1FF2h)
	+3F4h	Board Version(0x12)	(1FF4h)

Basically the memory allocation of the PCI-N404 board is same as that of BPHR with two modules installed, and the allocated memory area can be divided largely into two sections. One section is 1KB of memory space allocated for CAMC-QI (four axes), and the other section is the 2KB of memory space for checking the status of interrupt and board operation. Only the even address memories contain the valid values, i.e. the odd address memories are not used.

Table 3. Address map of access area of CAMC-QI

Offset	Name	Function	Length	R/W	Initial Value
000h	Preamble	PREAMBLE	8 Bit	R	B6h
002h	Sub ID	Sub function ID	8 Bit	R	06h
004h	Version	Hardware version	8 Bit	R	11h
200h	Axis 0 DATA PORT	CAMC-QI x axis data port LSB	16 Bit	W/R	00h
208h	Axis 0 DATA PORT	CAMC-QI x axis data port MSB	16 Bit	W/R	00h
210h	Axis 0 COMMAND PORT/ Axis 0 Main Status	CAMC-QI x axis command port(W) Main Status(R)	16 Bit	W/R	00h
218h	Axis 0 UIO PORT/ Axis 0 Sub Status	CAMC-QI x Universal In/Out/ Sub Status(12~15 bit)	16 Bit	W/R	-
220h	Axis 1 DATA PORT	CAMC-QI y axis data port LSB	16 Bit	W/R	00h
228h	Axis 1 DATA PORT	CAMC-QI y axis data port MSB	16 Bit	W/R	00h
230h	Axis 1 COMMAND PORT/ Axis 1 Main Status	CAMC-QI y axis command port(W) Main Status(R)	16 Bit	W/R	00h
238h	Axis 1 UIO PORT/ Axis 1 Sub Status	CAMC-QI y Universal In/Out/ Sub Status(12~15 bit)	16 Bit	W/R	-
240h	Axis 2 DATA PORT	CAMC-QI z axis data port LSB	16 Bit	W/R	00h
248h	Axis 2 DATA PORT	CAMC-QI z axis data port MSB	16 Bit	W/R	00h

250h	Axis 2 COMMAND PORT/ Axis 2 Main Status	CAMC-QI z axis command port(W) Main Status(R)	16 Bit	W/R	00h
258h	Axis 2 UIO PORT/ Axis 2 Sub Status	CAMC-QI z Universal In/Out/ Sub Status(12~15 bit)	16 Bit	W/R	-
260h	Axis 3 DATA PORT	CAMC-QI u axis data port LSB	16 Bit	W/R	00h
268h	Axis 3 DATA PORT	CAMC-QI u axis data port MSB	16 Bit	W/R	00h
270h	Axis 3 COMMAND PORT/ Axis 3 Main Status	CAMC-QI u axis command port(W) Main Status(R)	16 Bit	W/R	00h
278h	Axis 3 UIO PORT/ Axis 3 Status	CAMC-QI u Universal In/Out/ Sub Status(12~15 bit)	16 Bit	W/R	-



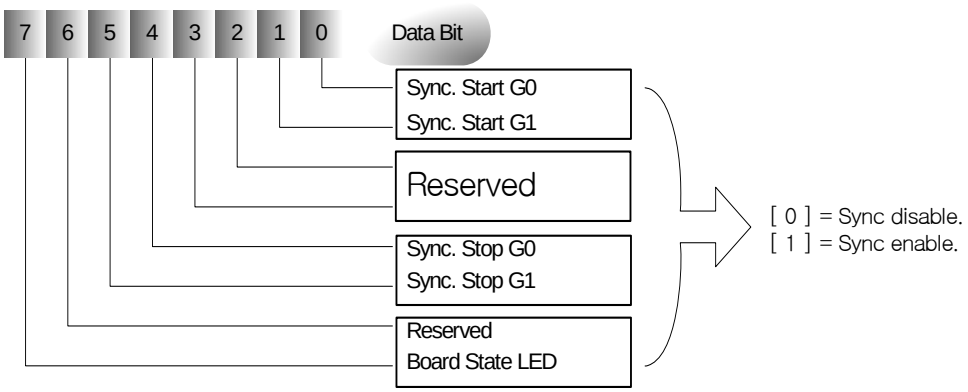
Picture 15. CAMC-QI Address Decoding of PCI-N404

3.4.2. Functional description of Registers

3.4.2.1. Board Configuration Area

1. SYNC: Read/Writable (1800h)

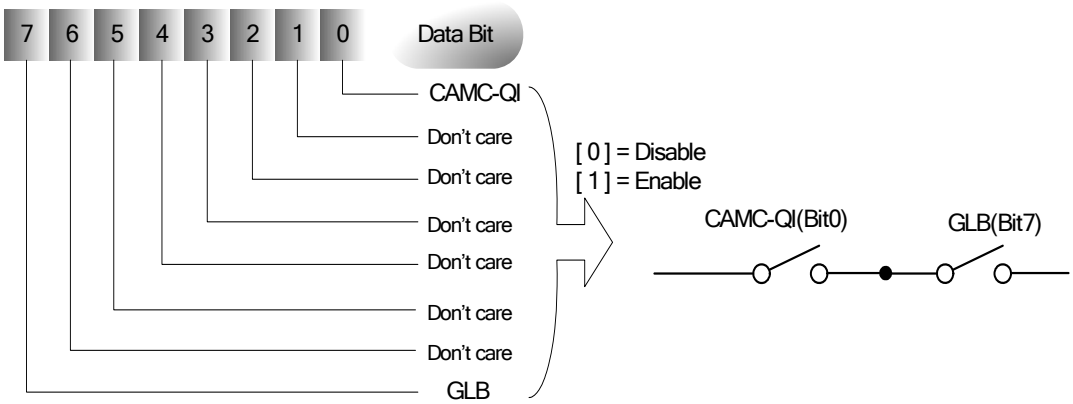
The sync port produces the sync start signal of CAMC-QI and synchronized drive is done by setting the bit corresponding to each axis to active (set to '1') when there is a need to start the motors of each axis simultaneously. Bit 0 and bit 1 are used as the signals for synchronized start and bit 4 and bit 5 are used as the signals for synchronized stop. Bit 7 can turn on or off the board status LED, and it is used for the user to check the current status of the board.



For example, in order to start the motors connected to the X axis (axis 0) and Y axis (axis 1) simultaneously, set the synchronized start conditions for axis 0 and axis 1 to the start of the drive triggered by the signal 1, and write xxxxxxx1 to the Sync Port.

2. Interrupt enable registers: Read/Writable (1804h)

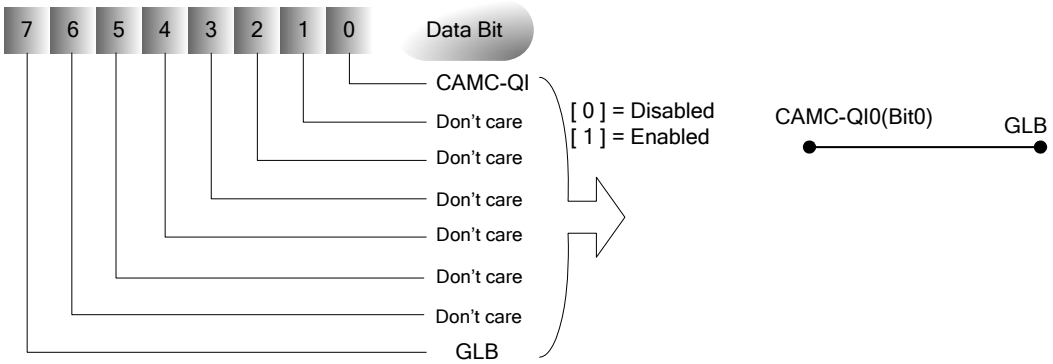
The interrupt enable register masks the interrupt request signals of CAMC-QI0 and CAMC-QI1, and setting a bit to '1' will enable the corresponding interrupt.



The GLB of bit7 is global enable/disable bit, and all interrupts will be disabled regardless of the configuration of other bits if this bit is set to '0', and setting this bit to '1' will enable or disable the interrupt request of CAMC-QI depending on the setting value of the bit 0.

3. Interrupt flag register: Read only (1806h)

The interrupt flag register indicates the status of the interrupt request signal of CAMC-QI, and bit value of '1' indicates that the CAMC-QI currently requests an interrupt. However, the interrupt request will be ignored if the interrupt is disabled for the corresponding port.



⑤ Preamble: Read only (1FF0h)

The Preamble register is a unique sequence of '0' and '1', and the value is set to '10110110' (0xB6). This register always contains the same value, and it can be used to identify the board of our company.

⑥ Board ID: Read only (1FF2h)

The board ID of the PCI-N404 board is '10000100' (0x84). Software libraries appropriate for each relevant ID must be used.

⑦ Board Version: Read only (1FF4h)

This register contains the information of board version for the purpose of identifying the hardware update status. The values for initial production version is '00010010'(0x12).

3.4.2.2. CAMC-QI Area

1. Preamble Register			
Register Offset	Register width	Write/Read	Initial Value
00h	8Bit	R	B6h

The initial value of the Preamble is B6h, and reading B6h from this register means the successful detection of CAMC-QI.

2. ID Register			
Register Offset	Register width	Write/Read	Initial Value
02h	8Bit	R	06h

ID is the indication of successful detection of CAMC-QI along with the Preamble.

3. Version Register			
Register Offset	Register width	Write/Read	Initial Value
04h	8Bit	R	10h

Version register is used to check the version information of CAMC-QI along with the ID.

4. CAMC-QI Chip Registers

Refer to the CAMC-QI chip manual for the details including the operation of each register and the instructions on the usage of commands.

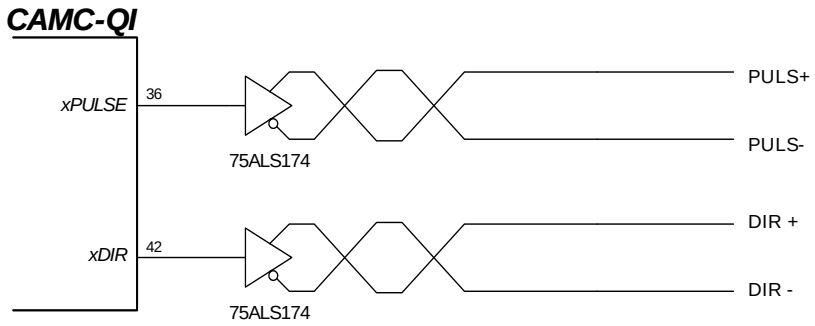
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3.5. Interface

3.5.1. Description of external interface pins

▶ Pulse output signal/direction output signal

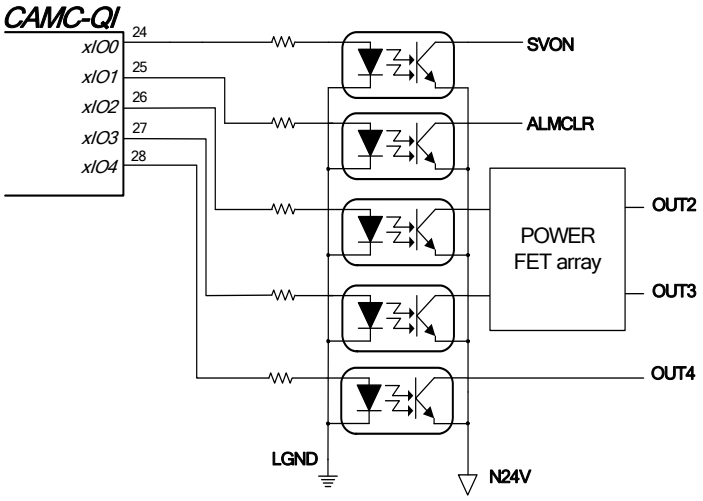
The pulse output signals and direction output signals from CAMC-QI chips are converted to the differential level signals by the line driver chip (75ALS174A) and connected to the external devices. The available pulse output modes are two-pulse mode, one-pulse mode, and two-phase pulse mode. When connecting to the pulse output signals and the direction output signals, which are line driver outputs, the twisted pair wires must be used to protect the signals from the external noise.



Picture 21. Interface of pulse output signals (X Axis)

▶ Servo ON, Alarm Clear signal, general purpose output signals (SVON, ALMC, OUT2, OUT3, OUT4)

There are total five of general purpose output signals in the PCI-N404 board using UIO0 ~ UIO4 of CAMC-QI. OUT0 of each axis is dedicated for SVON (Servo On), OUT1 is dedicated for ALMC (Alarm Clear), and OUT2~OUT4 are available for general purpose output. The maximum current of SVON/ALMC/OUT4 signals is 20 mA and the maximum current of OUT2~OUT3 is 100 mA. Users should be aware of the danger associated with the maximum current since exceeding the rated maximum current would cause damages of the board.



Picture 22. Interface of Servo ON, Alarm Clear, general purpose output signals (Axis X)

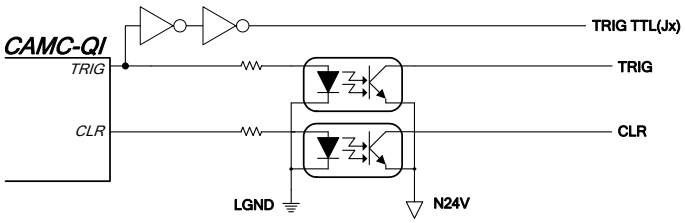
The output level of the board is inverted from the output level of the chip. That is, the output of the PCI-N404 board is N24V when the logic level of UOUT at the chip is '1', and the output of PCI-N404 is open collector when the logic level is '0'. All output signals are electrically isolated from the external signal by photo-couplers. The outputs are normally-open type, therefore users must apply the pull-up resistors in order to get the voltage of 24V level. (About 10㏀ resistor is used for pull-up resistor)

⚠	Care must be taken not to exceed the maximum current of 20 mA (±10%) for Servo On, Alarm clear, and OUT4 outputs. Exceeding the maximum current may damage the board.
⚠	Care must be taken not to exceed the maximum current of 100 mA (±10%) for OUT2 and OUT3 outputs. Exceeding the maximum current may damage the board.

▶ TRIG, CLR output signals

PCI-N404 has a function to generate the trigger and interrupt signals at the specified positions. And also there is a function that removes the surplus pulses of the servo motor drive, and the details of the trigger and clear signals are described here. The output level is inverted from the chip output level. That is, the TRIG and CLR output are N24V when their corresponding logic levels are '1', and open collector when the logic levels are '0'. Both signals are electrically isolated form the external signals by photo-couplers as general purpose output signals. These signals are generated at the conditions of specific positions or other conditions, and there will be a time delay is due to the delay of photo-coupler. CLR signal triggered by PELM and NELM sensors is delayed by 6㎲~8㎲, and the delay of 4㎲~5㎲ is applied from the specified position

where the TRIG signal should be generated until the TRGI signal is generated. The small amount of delay would impose a significant limitation when the trigger signal has to be generated at a specified position during the high speed operation. In this case, it is recommended to use TTL level TRIG signal in J3/5 of the PCI-404. The output of the photo-coupler is normally open type, therefore users must apply the pull-up resistor in order to get the voltage of 24V level. Use the resistor of about 10㏀ for pull-up resistor.

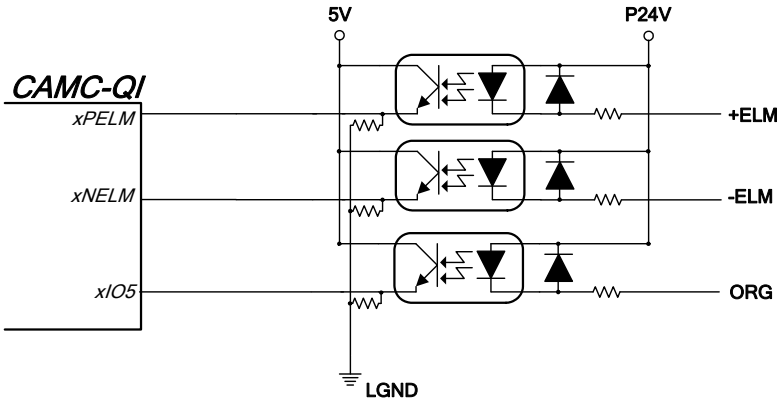


Picture 23. Trigger signals generated at specified positions and clear signal output for removing surplus pulses

⚠	Care must be taken not to exceed the maximum current of 20 mA (±10%) for the TRIG and CLR output. Exceeding the maximum current may damage the board.

▶ Limit sensor signals and home sensor signals (+ELM, -ELM, ORG)

The role of limit sensor signals is to stop the forward or backward driving pulses when exceeding the limits. The I05 input terminal of CAMC-QI of each axis is for connecting with the home sensor that is for searching for home position.



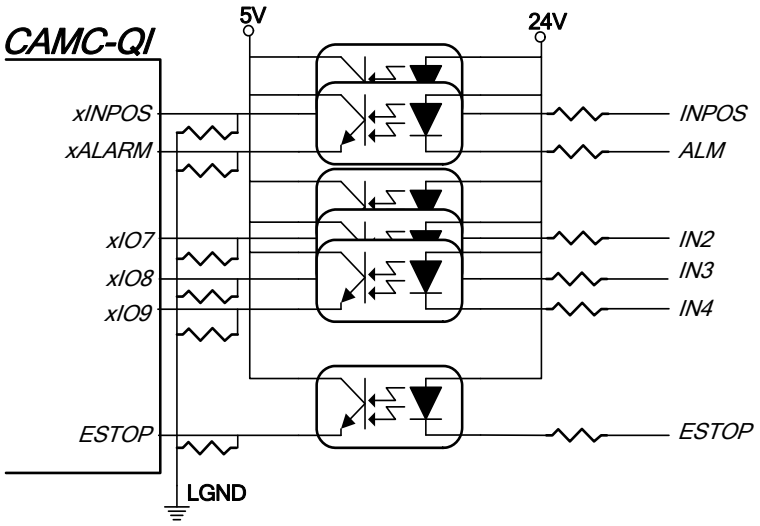
Picture 24. Interfaces for the signals of LIMIT +/- and home sensor (Axis X)

Signals for the limit sensors go through the photo-couplers for electrical isolation and the Schmitt trigger circuit and digital filters built into the input stage of the CAMC-QI reduces the interference from the external noise. External power source of +24V is required to get the limit

trigger input. The threshold of limit signal detection can be programmed inside the CAMC-QI chip. The signals from the home sensor also go through the photo-coupler for electrical isolation, and digital filter and Schmitt trigger circuit for removing the effects of noise. The logic level seen at the chip is '0' when the sensor output is open or P24V, and '0' when the sensor output is N24V. The response time to these signals is 10 ~20uSec.

▶ INPOS, ALM, general purpose input 2, 3, 4 signals, other sensor signal inputs (INPOS, ALM, IN2, IN3, IN4, STOP)

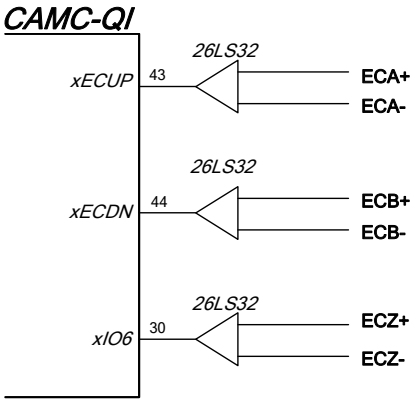
INPOS signal is generated from the servo drive to determine the completion of the position drive when the position specified in the velocity profile is reached. ALM signal is activated when the servo drive in error state. General purpose input signals are available for general applications for users. EMGN, SSTOP signals are for soft stop and emergency stop and these signals can be connected to the emergency switch or the output of other devices. These signals are electrically isolated by photo-couplers in the interface circuits.



Picture 25. Interface of INPOS, ALM, IN2, IN3, EMGN, SSTOP(Axis X)

▶ Encoder signals (ECA +/-, ECB +/-, ECZ +/-)

ECA, ECB signals are used external input signals to the counter registers of CAMC-QI, connected to CAMC-QI chips by line receivers (26LS32).

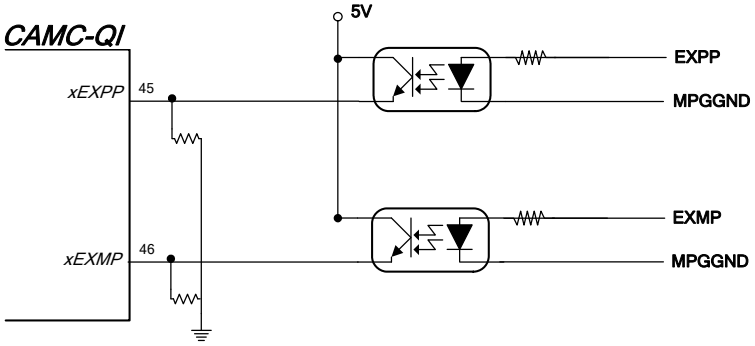


Picture 26. Interface of A, B, Z-phase encoder signals

The Z-phase of the encoder (INDEX) is used for home search and is connected to UIO6 of CAMC-QI. Encoder signals, which are differential input signals, require twisted pair cables to connect to external encoders in order to remove external noise. Line receivers used in PCI-N404 can process up to 4㎒ of signals, and are capable of 16MCount/Sec in two-phase quadruple multiplication mode.

▶ MPG input signals (EXPP, EXMP)

EXPP and EXMP signals are used as input signals to CAMC-QI from external rotary encoders and these signals are isolated from CAMC-QI device with high-speed photo-couplers that support up to 1.2MHz.

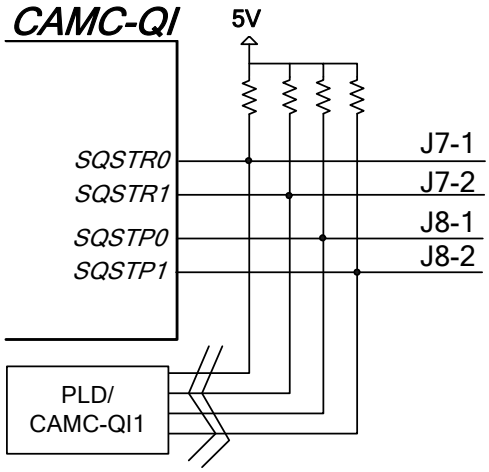


Picture 27. Interface circuit of differential MPG input signals

Each MPG input signal is shared between two axes, i.e. the signal is not used for one axis only. The MPG signal input to the axis 0 is also connected to the MPG input of the axis 2, and the signal input to the axis 1 is also connected to the MPG input of the axis 3. The MPG signal input pins for the axis 2 and 3 in the J2 external connector are not used.

▶ Dynamic Link signal

SQSTR1/2 input and SQSTP1/2 input of CAMC-QI are used as dynamically linked sync start, and sync stop signal, respectively.



Picture 27. Interface of Dynamic link input signal

SQSTR0/1and SQSTP0/1 of CAMC-QI are to be used in order to use the synchronization function, and the scheduled synchronized motor drive can be started/stopped when one or more inputs (external input through J7/J8, PLD output, CAMC-QI output) are in logical low state, which are in high impedance (High-Z) state when not used.

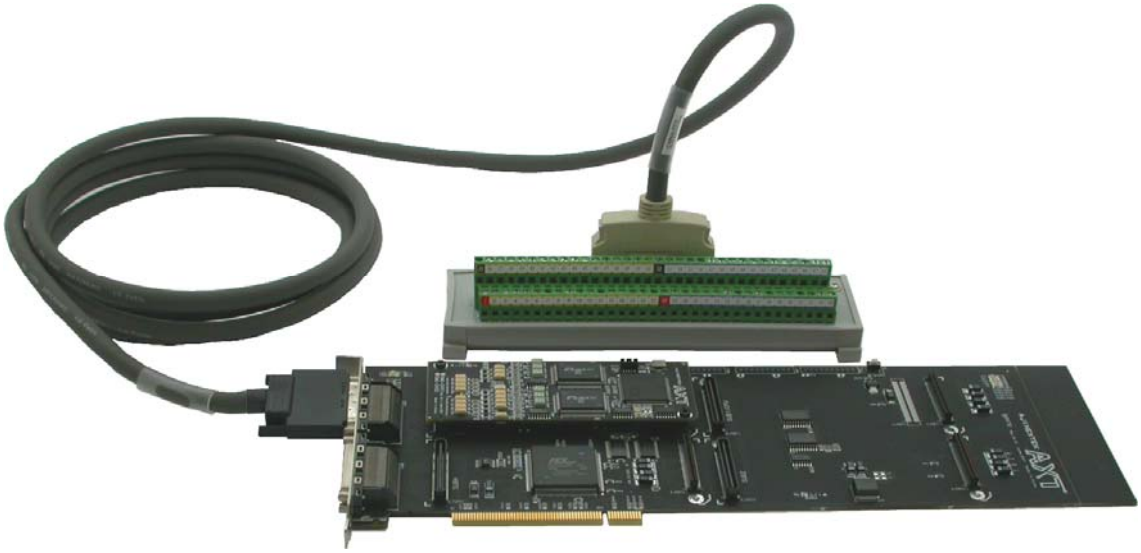
▶ Logic Ground (LGND)

There might be a need to connect the ground signal depending on the type of servo drive or encoder. PCI-N404 provides one logic ground (LGND) for each axis for this purpose.

⚠	Care must be taken since the connection of the noise source or power source to the ground pin may cause serious damage to the control system.

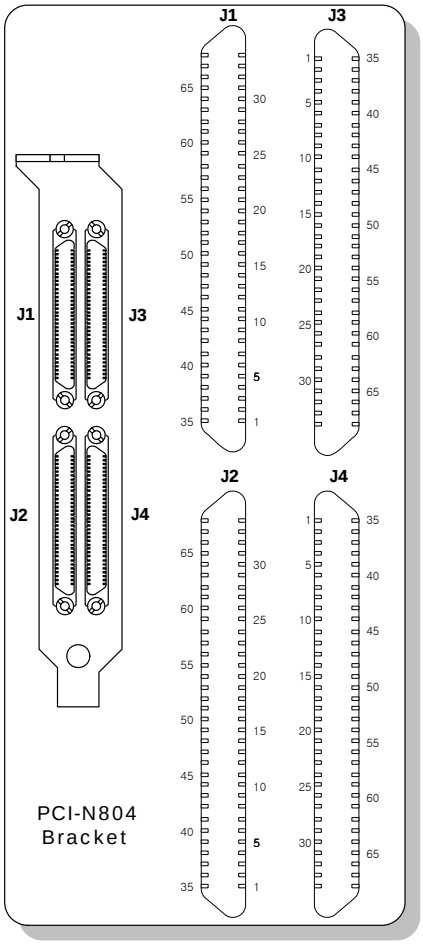
4. Board Connection

4.1. Connection with Terminal Block



Picture 5. Picture of connection between PCI-N404 board and terminal block

4.2. Description of pins for external connection



Picture 28. Connectors placement of the board

Product name of the Board J1 ~ J4 : AMP-787962-1. (Receptacle)
Product name of the Connector(Plug) : Honda HDRA-E68MA1

Table 4. Pin description of Board J1 ~ J2 (J3~J4 : No connection) Connectors

J1	Name	I/O	Pin Description	J1	Name	I/O	Pin Description
1	X_P24V	I	External power for IO drive	35	X_N24V	I	External power ground for IO drive
2	X_PULS+	O	Pulse output signal (LineDriver+)	36	X_PULS-	O	Pulse output signal (LineDriver-)
3	X_DIR+	O	Direction output signal (LineDriver	37	X_DIR-	O	Direction output signal (LineDriver
4	X_SVON	O	SERVO-ON output	38	X_INPO	I	Inposition signal for servo driver
5	X_ALM	I	Alarm signal of servo drive	39	X_ALMC	O	Alarm clear signal output
6	LGND	O	Logic GND	40	X_CLR	O	Output signal to clear surplus pulses
7	X_ECA+	I	Encoder A+ input	41	X_ECA-	I	Encoder A- input
8	X_ECB+	I	Encoder B+ input	42	X_ECB-	I	Encoder B- input
9	X_ECZ+	I	Encoder Z+ input	43	X_ECZ-	I	Encoder Z- input
10	X_IN2	I	General purpose input 2	44	X_OUT2	O	General purpose output 2
11	X_IN3	I	General purpose input 3	45	X_OUT3	O	General purpose output 3
12	X_IN4	I	General purpose input 4	46	X_OUT4	O	General purpose output 4
13			No connected	47			No connected
14	X_EXPP	I	MPG EXPP(+) input	48	X_EXMP	I	MPG EXMP(+) input
15	X_MPG-	I	MPG EXMP/EXPP(-)	49	X_+ELM	I	Forward direction limit sensor inpu
16	X_-ELM	I	backward direction limit sensor input	50	X_ORG	I	Home sensor signal input
17	X_STOP	I	Emergency stop switch (sensor) input	51	X_TRIG	O	Trigger output signal (at specified positions)
18	Y_P24V	I	External power for IO drive	52	Y_N24V	I	External power ground for IO drive
19	Y_PULS+	O	Pulse output signal (LineDriver+)	53	Y_PULS-	O	Pulse output signal (LineDriver-)
20	Y_DIR+	O	Direction output signal (LineDriver	54	Y_DIR-	O	Direction output signal (LineDriver
21	Y_SVON	O	SERVO-ON output	55	Y_INPO	I	Inposition signal for servo driver
22	Y_ALM	I	Alarm signal of servo drive	56	Y_ALMC	O	Alarm clear signal output
23	LGND	O	Logic GND	57	Y_CLR	O	Output signal to clear surplus Pulses
24	Y_ECA+	I	Encoder A+ input	58	Y_ECA-	I	Encoder A- input
25	Y_ECB+	I	Encoder B+ input	59	Y_ECB-	I	Encoder B- input
26	Y_ECZ+	I	Encoder Z+ input	60	Y_ECZ-	I	Encoder Z- input
27	Y_IN2	I	General purpose input 2	61	Y_OUT2	O	General purpose output 2
28	Y_IN3	I	General purpose input 3	62	Y_OUT3	O	General purpose output 3
29	Y_IN4	I	General purpose input 4	63	Y_OUT4	O	General purpose output 4
30			No connected	64			No connected
31	Y_EXPP	I	MPG EXPP(+) input	65	Y_EXMP	I	MPG EXMP(+) input
32	Y_MPG-	I	MPG EXMP/EXPP(-)	66	Y_+ELM	I	Forward direction limit sensor inpu
33	Y_-ELM	I	backward direction limit sensor input	67	Y_ORG	I	Home sensor signal input
34	Y_STOP	I	Emergency stop switch (sensor) input	68	Y_TRIG	O	Trigger output signal (at specified positions)

J2	Name	I/O	Pin Description	J2	Name	I/O	Pin Description
1	Z_P24V	I	External power for IO drive	35	Z_N24V	I	External power ground for IO drive
2	Z_PULS+	O	Pulse output signal (LineDriver+)	36	Z_PULS-	O	Pulse output signal (LineDriver-)
3	Z_DIR+	O	Direction output signal (LineDriver	37	Z_DIR-	O	Direction output signal (LineDriver
4	Z_SVON	O	SERVO-ON output	38	Z_INPO	I	Inposition signal for servo driver
5	Z_ALM	I	Alarm signal of servo drive	39	Z_ALMC	O	Alarm clear signal output
6	LGND	O	Logic GND	40	Z_CLR	O	Output signal to clear surplus Pulses
7	Z_ECA+	I	Encoder A+ input	41	Z_ECA-	I	Encoder A- input
8	Z_ECB+	I	Encoder B+ input	42	Z_ECB-	I	Encoder B- input
9	Z_ECZ+	I	Encoder Z+ input	43	Z_ECZ-	I	Encoder Z- input
10	Z_IN2	I	General purpose input 2	44	Z_OUT2	O	General purpose output 2
11	Z_IN3	I	General purpose input 3	45	Z_OUT3	O	General purpose output 3
12	Z_IN4	I	General purpose input 4	46	Z_OUT4	O	General purpose output 4
13			No connected	47			No connected
14			No connected	48			No connected
15			No connected	49	Z_+ELM	I	Forward direction limit sensor inpu
16	Z_-ELM	I	backward direction limit sensor input	50	Z_ORG	I	Home sensor signal input
17	Z_STOP	I	Emergency stop switch (sensor) input	51	Z_TRIG	O	Trigger output signal (at specified positions)
18	U_P24V	I	External power for IO drive	52	U_N24V	I	External power ground for IO drive
19	U_PULS+	O	Pulse output signal (LineDriver+)	53	U_PULS-	O	Pulse output signal (LineDriver-)
20	U_DIR+	O	Direction output signal (LineDriver	54	U_DIR-	O	Direction output signal (LineDriver
21	U_SVON	O	SERVO-ON output	55	U_INPO	I	Inposition signal for servo driver
22	U_ALM	I	Alarm signal of servo drive	56	U_ALMC	O	Alarm clear signal output
23	LGND	O	Logic GND	57	U_CLR	O	Output signal to clear surplus Pulses
24	U_ECA+	I	Encoder A+ input	58	U_ECA-	I	Encoder A- input
25	U_ECB+	I	Encoder B+ input	59	U_ECB-	I	Encoder B- input
26	U_ECZ+	I	Encoder Z+ input	60	U_ECZ-	I	Encoder Z- input
27	U_IN2	I	General purpose input 2	61	U_OUT2	O	General purpose output 2
28	U_IN3	I	General purpose input 3	62	U_OUT3	O	General purpose output 3
29	U_IN4	I	General purpose input 4	63	U_OUT4	O	General purpose output 4
30			No connected	64			No connected
31			No connected	65			No connected
32			No connected	66	U_+ELM	I	Forward direction limit sensor inpu
33	U_-ELM	I	backward direction limit sensor input	67	U_ORG	I	Home sensor signal input
34	U_STOP	I	Emergency stop switch (sensor) input	68	U_TRIG	O	Trigger output signal (at specified positions)

4.3. Terminal Block

PCI-N404 requires the terminal blocks to connect the board with the motor drives and sensors. Provided terminal blocks are 68 pins of either screw fixed type or detachable type. Terminal block can be mounted on DAIN rail and comprises cable connector and terminals contacts for connecting with servo drives and sensors. Pin map of the terminal block is identical to that of the carrier board, but caution is needed since the pin array of the terminal block is different from the pin map of the connector.

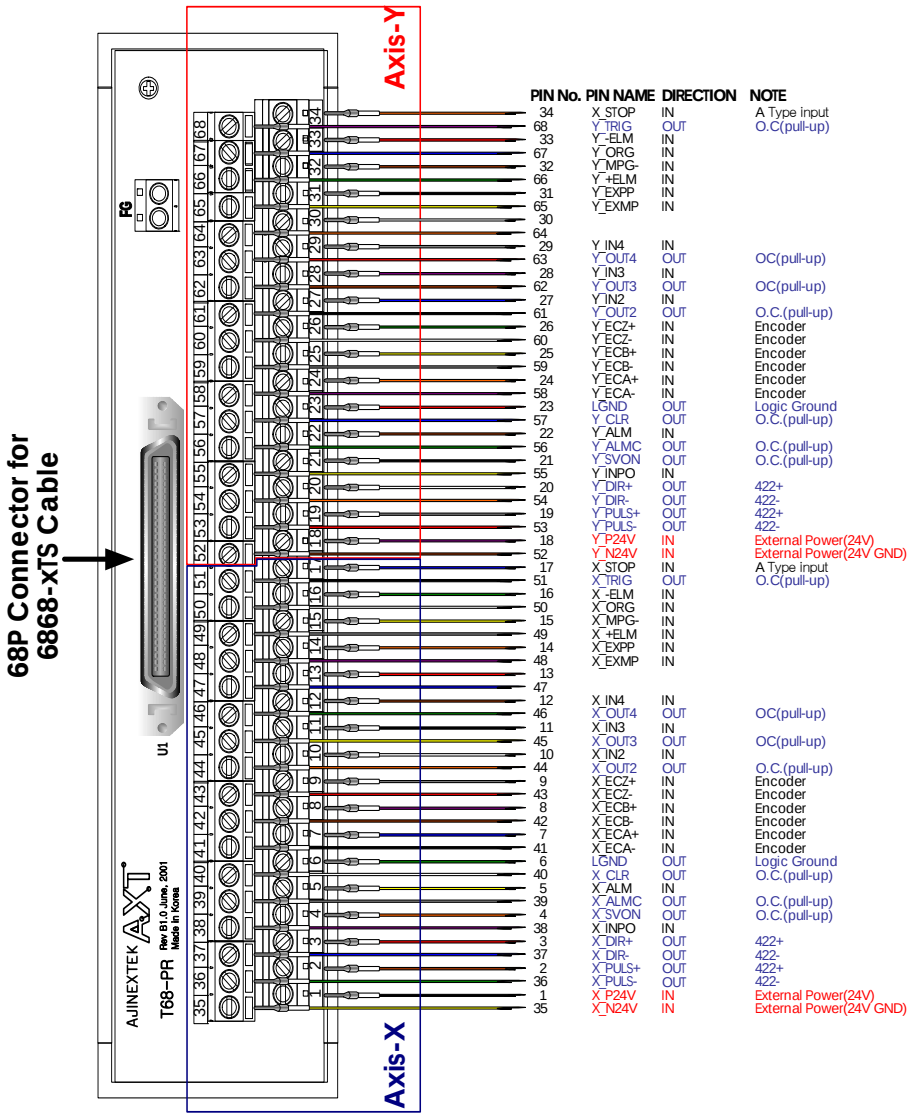
Table 19. Terminal block specification

Product name	Height	Length	Width	Terminal Pitch	Fix Method	Wiring Method	Cable
T68-PR v1.0	43.0mm	200.5mm	52.0mm	5.0mm	Screw	Fixed	C6868-xTS
T68-PR v2.0	57.0mm	200.5mm	52.0mm	5.0mm	Screw	Removable	C6868-xTS



Picture 29. Product photograph of T68-PR v1.0 terminal block (top) and T68-PR v2.0 terminal block (bottom)

The picture shown below is the connection diagram of the T68-PR terminal block that shows the connection of the PCI-N404 board, servo drives or step drives, and other sensors. The wiring might change for different servo drives, and the names of signals and the markings in the terminals may not be identical. The connection diagram for various models of servo drives and step drives from the various manufacturers are provided in the application notes for your reference.



Picture 6. Connection diagram of T68-PR v1.0 and T68-PR v2.0

Table 20. Description of T68-PR v1.0 & T68-PR v2.0 terminal block pin functions

Pin No.	Pin Name	Pin Description	Pin No.	Pin Nam	Pin Description
1	X_P24V	External power for IO drive	35	X_N24V	External power ground for IO drive
2	X_PULS+	Pulse output signal (LineDrive	36	X_PULS-	Pulse output signal (LineDriver-)
3	X_DIR+	Direction output si (LineDriver+)	37	X_DIR-	Direction output signal (LineDriver
4	X_SVON	SERVO-ON output	38	X_INPO	Inposition signal for servo driver
5	X_ALM	Alarm signal of servo drive	39	X_ALMC	Alarm clear signal output
6	LGND	Logic GND	40	X_CLR	Output signal to clear surplus Pulses
7	X_ECA+	Encoder A+ input	41	X_ECA-	Encoder A- input
8	X_ECB+	Encoder B+ input	42	X_ECB-	Encoder B- input
9	X_ECZ+	Encoder Z+ input	43	X_ECZ-	Encoder Z- input
10	X_IN2	General purpose input 2	44	X_OUT2	General purpose output 2
11	X_IN3	General purpose input 3	45	X_OUT3	General purpose output 3
12	X_IN4	General purpose input 4	46	X_OUT4	General purpose output 4
13		No connected	47		No connected
14	X_EXPP	MPG EXPP(+) input	48	X_EXMP	MPG EXMP(+) input
15	X_MPG-	MPG EXMP/EXPP(-)	49	X_+ELM	Forward direction limit sensor inpu
16	X_-ELM	backward direction limit sensor input	50	X_ORG	Home sensor signal input
17	X_STOP	Emergency stop switch(sen input	51	X_TRIG	Trigger output signal (at specified positions)
18	Y_P24V	External power for IO drive	52	Y_N24V	External power ground for IO drive
19	Y_PULS+	Pulse output signal (LineDrive	53	Y_PULS-	Pulse output signal (LineDriver-)
20	Y_DIR+	Direction output si (LineDriver+)	54	Y_DIR-	Direction output signal (LineDriver
21	Y_SVON	SERVO-ON output	55	Y_INPO	Inposition signal for servo driver
22	Y_ALM	Alarm signal of servo drive	56	Y_ALMC	Alarm clear signal output
23	LGND	Logic GND	57	Y_CLR	Output signal to clear surplus Pulses
24	Y_ECA+	Encoder A+ input	58	Y_ECA-	Encoder A- input
25	Y_ECB+	Encoder B+ input	59	Y_ECB-	Encoder B- input
26	Y_ECZ+	Encoder Z+ input	60	Y_ECZ-	Encoder Z- input
27	Y_IN2	General purpose input 2	61	Y_OUT2	General purpose output 2
28	Y_IN3	General purpose input 3	62	Y_OUT3	General purpose output 3
29	Y_IN4	General purpose input 4	63	Y_OUT4	General purpose output 4
30		No connected	64		No connected
31	Y_EXPP	MPG EXPP(+) input	65	Y_EXMP	MPG EXMP(+) input
32	Y_MPG-	MPG EXMP/EXPP(-)	66	Y_+ELM	Forward direction limit sensor inpu
33	Y_-ELM	backward direction limit sensor input	67	Y_ORG	Home sensor signal input
34	Y_STOP	Emergency stop switch (sensor) input	68	Y_TRIG	Trigger output signal (at specified positions)

4.4. Cable

The C6868-xTS cable is to connect the PCI-N404 board and T68-PR terminal block. This cable consists of 34 twisted pairs from 68 individual wires, and the entire cable assembly is shielded. The cable is rated for 150 mA per each pin so that motor drives or sensors can be directly connected.



Picture 7. Product photo of C6868-3TS

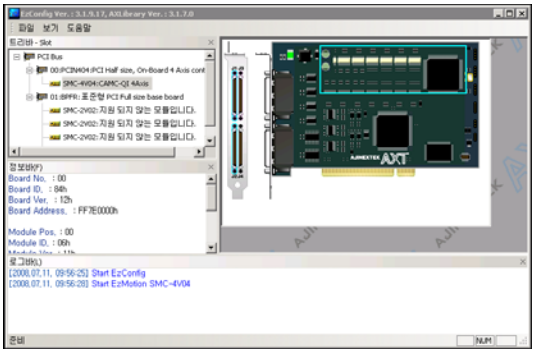
Table 5. C6868-xTS specifics (x = length[m])

Product Name	Current capacity	No. of pins	Pitch	Cable Type	Length(x)	Applicable Terminal Block
C6868-xTS	150mA (per pin)	68pin	0.8mm – 1.27mm (asymmetric)	Twist Pair	1m, 2m, 3m …	T68-PR v1.0 T68-PR v2.0

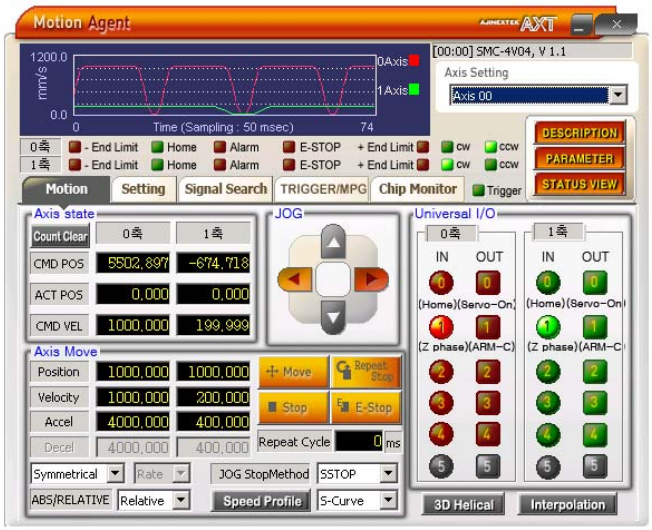
4.5. Agent

Install the software by referring to the software installation guide after the installation of the hardware with the installation instructions and hardware installation guide. EzSoftware to be installed includes EzConfig that provides the management function of the AXT products installed in the system, EzAgent for testing each function and module, Libraries that user program will use, and etc. Start the EzMotion Agent by referring to the EzConfig manual and the EzMotion Agent manual after verifying that the software is installed correctly.

EzMotion is the Agent tools for Windows for PCI-N404. And the following pictures are the screen shot of the EzConfig (top, picture 9) and EzMotion (bottom, picture 9).



EzConfig screen that shows the result of board search



Execution of EzMotion Agent

Picture 9. Screen snapshot of the EzConfig and EzMotion Agent Software

5. Order Information for Related Products

5.1. Terminal Block

Table 25. Terminal Block Types

Product Nam	Height	Length	Width	Terminal Pitch	Fix Metho	Terminal Type	Applicable AnyMotion produc
T36-PR v1.0	43.0mm	128.0mm	52.0mm	5.0mm	Screw	Fixed	SMC-1V01 SMC-1V02
T36-PR v2.0	57.0mm	128.0mm	52.0mm	5.0mm	Screw	Detachable	SMC-1V01 SMC-1V02
T68-PR v1.0	43.0mm	200.5mm	52.0mm	5.0mm	Screw	Fixed	SMC-2V01 SMC-2V02 SMC-2V03 SMC-4V51(2) SMC-4V52(2) SMC-2V53 PCI-N404 PCI-N804
T68-PR v2.0	57.0mm	200.5mm	52.0mm	5.0mm	Screw	Detachable	SMC-2V01 SMC-2V02 SMC-2V03 SMC-4V51(2) SMC-4V52(2) SMC-2V53 PCI-N404 PCI-N804

5.2. Cable

Table 26. Cable Types

Product Name	Current Capacity	Number of pins	Pitch	Cable Type	Length	Product name of applicab motion modules
C6836-xTS	150mA	36pin	0.8mm – .27mm(asymmetric)	Twist Pair	1m, 2m, 3m, …	SMC-1V01 SMC-1V02
C6868-xTS	150mA	68pin	0.8mm – .27mm(asymmetric)	Twist Pair	1m, 2m, 3m, …	SMC-2V01 SMC-2V02 SMC-2V03 SMC-4V51(2) SMC-4V52(2) SMC-2V53 PCI-N404 PCI-N804



Refer to the product selection guide for the terminal blocks and cables. Once the terminal block is decided, then the cables to connect the board and the terminal blocks will be decided accordingly. That is, C6836-xTS cable must be used for T36-PR terminal block, and C6868-xTS cable must be used for the terminal block T68-PR. The required number of terminal blocks and cables are determined by the number of axis. Two axes require one terminal and one cable, and four axes board requires two pairs of cable and terminal block.

5.3. Definition of Terms :

AnyBus :

Common designation of the carrier board of the Ajinextek that support PCI, ISA, CPCI, and VME BUS

AnyMotion :

Common designation of motion control modules of Ajinextek that provide the control function for various step motors, servo motors, and etc

AnyDIO :

Common designation of digital input/output control modules of Ajinextek that provide the functions to connect with various sensors

AnyAIO :

Common designation of analog input/output control modules of Ajinextek that provide the functions to connect with various sensors

AnyCOM :

Common designation of the communication module of Ajinextek that provide the communication function

AXL:

Integrated AjineXtek Library

EzConfig, EzMotion, EzDIO, EzAI, EzAO, EzCOM Agent :

Configuration and operation support software tools of Ajinextek for each of AnyMotion, AnyDIO, AnyAIO, AnyCOM modules installed in the AnyBus carrier board.

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